1.Git-HOL

Git is a version control system that helps us track changes in our code. It allows us to save different versions of our project, work safely on changes, and collaborate with others using platforms like GitHub.
In this exercise, I learned the basic Git workflow and understood the purpose of commands like git init, git status, git add, git commit, git push, and git pull.

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1. git init
git init is used to create a new Git repository in a project folder.
When we run this command, Git starts tracking the changes in that folder and creates a hidden .git folder that stores all the information related to version control.
Command:
git init
Example:
If I create a new project folder and run git init, that folder becomes a Git repository where I can track my code changes.

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2. git status
git status is used to check the current state of the repository.
It shows:

- Which files are modified
- Which files are added to the staging area
- Which files are not tracked by Git
- The current branch information

Command:
git status
Example:
After creating a new file, git status shows that the file is untracked and needs to be added before committing.

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3. git add
git add is used to move files from the working area to the staging area.
The staging area is a place where we prepare the changes that we want to save in the next commit.
Command:
git add filename
To add all files:
git add .

Example:
If I create index.html and want Git to track it, I use:
git add index.html
Now the file is ready to be committed.

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4. git commit
git commit is used to permanently save the changes that are present in the staging area.
A commit creates a checkpoint of the project at a particular point in time.
Command:
git commit -m "commit message"
Example:
git commit -m "Added homepage design"
The commit message explains what changes were made, making it easier to understand the project history.

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5. git push
git push is used to upload the committed changes from the local repository to a remote repository like GitHub.
It helps us store our code online and share it with others.
Command:
git push origin main
Example:
After committing my changes locally, I use git push to upload those changes to my GitHub repository.

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6. git pull
git pull is used to download the latest changes from a remote repository and update the local repository.
It is mainly used when working with a team where other members may have pushed new changes.
Command:
git pull origin main
Example:
If a teammate updates the project on GitHub, I can use git pull to get those latest changes on my local machine.

Git Workflow Understanding
The basic Git workflow that I learned is:
Create Project
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git init

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Make Changes

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git status

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git add

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git commit

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git push

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git pull (to get latest updates)